PROMPT FOR RESOLVING THE PROJECTIVE PLANE OF ORDER 12

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Abstract. This document contains a full frontier-model research prompt for deciding whether a finite projective plane of order 12 exists. A positive answer is a 157×157 incidence matrix; a negative answer requires an exhaustive, independently checkable nonexistence proof. Status snapshot: 20 July 2026.

1. Prompt

Current task statement

A finite projective plane of order q has $q^2 + q + 1$ points and the same number of lines; every line contains $q + 1$ points, every point lies on $q + 1$ lines, every two points determine exactly one line, and every two lines meet in exactly one point.

Resolve the order-12 case completely. Such a plane would have 157 points and 157 lines, with 13 incidences per point and per line.

Equivalently, decide whether there exists a binary incidence matrix $N \in \{0, 1\}^{157 \times 157}$ satisfying

$$N\mathbf{1} = 13\mathbf{1}, \qquad N^{\mathsf{T}}\mathbf{1} = 13\mathbf{1}, \qquad NN^{\mathsf{T}} = 12I + J.$$

Assume for purposes of this task that a complete resolution is reachable. Do not assume existence or nonexistence.

Complete-resolution standard

A complete solution must prove one and only one of the following:

- Positive alternative: provide a complete incidence structure satisfying every projective-plane axiom exactly;
- Negative alternative: prove that no such incidence structure exists, either conceptually or through a formally justified exhaustive computation;
- for computational work, prove that every tactical decomposition, prescribed subconfiguration, automorphism case, and canonical augmentation step covers the full space.

What does not count

Partial progress is insufficient unless it logically implies the exact resolution above. In particular, the following do not count:

- a partial plane, affine plane fragment, block design with incorrect pair intersections, or incidence matrix satisfying only row and column sums;
- a Bruck-Ryser-Chowla calculation that does not actually exclude order 12;
- excluding planes with a chosen collineation group while leaving the trivial-automorphism case untreated;
- a balanced multi-splittable Hadamard candidate without a proved equivalence and complete parameter verification;
- a SAT or exact-cover failure without a complete certificate and a proof of encoding completeness;
- computational verification under an unproved normal form or through any bounded subset of possible planes.

Use multiagent v2 aggressively and dynamically. You have up to 64 concurrent agents available. Do not use a fixed assignment such as "N agents for strategy X." Instead, manage the search using the following heuristics:

- Begin with a genuinely diverse portfolio of approaches. Agents should explore substantially different formulations, invariants, reductions, algebraic viewpoints, structural inductions, decompositions, optimization methods, exact computational searches, and formal-proof routes.
- Do not tell most agents the currently favored approach. Preserve independence during early rounds so that agents do not all converge to the same attractive but incomplete reduction.
- Maintain an explicit registry of approach families. Group agents by the mathematical mechanism they are using, not by superficial wording. If too many agents converge to one family, redirect some toward underexplored formulations.
- Do not allow one route to dominate merely because it gives elegant reductions. A route that ends at a missing lemma equivalent in strength to the original problem is not close to completion unless it supplies a genuinely new proof of that lemma.
- When an approach stalls at a theorem-strength missing lemma, mark that route as blocked. Reopen it only when an agent proposes a materially new mechanism, invariant, construction, or exact experiment.
- Keep several incompatible proof or construction routes alive through multiple rounds. Cross-pollinate ideas only after independent agents have developed them far enough to expose their real strengths and gaps.
- Use adversarial agents throughout. Every candidate theorem, construction, certificate, reduction, and program must be attacked independently. Assign separate agents to mathematical correctness, completeness of case analysis, numerical exactness, implementation soundness, and novelty checking.
- Require agents to return concrete lemmas, constructions, equations, counterexamples to proposed sublemmas, executable encodings, or formally stated proof obligations. Reject status reports, vague optimism, and claims that a global compatibility step is "routine."
- The root agent should repeatedly synthesize, challenge, redirect, and launch new rounds. Do not stop after the first wave fails. Convert promising patterns into exact conjectures and send those conjectures to independent proof and falsification teams.

Problem-specific search portfolio

In the initial independent rounds, ensure that the portfolio includes, but is not limited to, the following genuinely different families:

- direct incidence-matrix SAT, exact-cover, CP-SAT, pseudo-Boolean, or integer-programming formulations with proof-producing solvers;
- canonical augmentation from partial linear spaces, with strong isomorph rejection and mathematically proved extension rules;
- tactical decompositions and the very restricted possible collineation groups, while maintaining a complete branch for trivial automorphism group;
- coding theory from the row span over $\mathbb{F}_2$, $\mathbb{F}_3$, and other characteristics, including weight enumerators and divisibility constraints;
- balancedly multi-splittable Hadamard matrices of order 144 and other exact equivalences with designs or orthogonal arrays;
- Baer subplanes, arcs, blocking sets, ovals, unitals, spreads, and other forced or forbidden subconfigurations;
- p-rank, Smith-normal-form, modular, determinant, and representation-theoretic obstructions that apply to every possible incidence matrix;
- construction search through difference sets, quasifields, planar ternary rings, coordinatization, loops, and non-Desarguesian algebraic systems.

Mandatory adversarial audit

Any surviving candidate must be checked against every item below by agents that did not originate the candidate:

- check all 157^2 entries are binary and that all row and column sums are exactly 13;
- check every pair of distinct points lies on exactly one common line and every pair of distinct lines meets in exactly one point;

- verify $NN^{\mathsf{T}} = 12I + J$ and, independently, $N^{\mathsf{T}}N = 12I + J$;
- ensure point and line labels are distinct and no duplicate blocks are hidden by file formatting;
- if using equivalence to another object, formalize both directions and verify all side conditions at $q = 12$;
- if claiming nonexistence, audit the completeness of automorphism cases, tactical decompositions, canonical labels, and all solver certificates;
- test the implementation on known planes of prime-power order and on the known nonexistence proof at order 10 to expose encoding errors.

Required final artifact

The final response must contain the mathematical proof and a complete archival package:

- for existence, the full incidence matrix, point-line lists, hashes, and at least two independent exact verifiers;
- for nonexistence, the reduction theorem, complete case tree, solver inputs, proof certificates, checkers, and case-count audit;
- a concise proof that the machine representation is equivalent to the projective-plane axioms;
- commands and software versions sufficient for independent reproduction from raw source files.

Public search may be used for standard finite-geometry theory, known restrictions on order 12, and prior computational methodology. Do not search for a claimed unpublished plane or merely answer that order 12 is the first unresolved case.

Do not return merely because current approaches fail or agents report theorem-strength gaps. Continue launching new rounds, reopening blocked approaches only when there is a genuinely new mechanism, and searching for fresh formulations.

Return only when a complete resolution has been found and survives adversarial audit. Do not return a reduction, a special case, an isolated missing lemma, an improved bound that leaves the target unresolved, a "best effort" literature review, or an explanation of why the problem is difficult.

Spend at least 8 hours of sustained search before even considering return or termination. If the execution environment imposes a hard stop before a complete resolution, do not fabricate success or elapsed work: preserve all machine-checkable artifacts and report only the strongest rigorously verified derivation together with the exact remaining gap.